Kuan-Chen Chen

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EDUCATION

University of Florida **Gainesville, FL**
Master of Science in Applied Data Science *8/2024 - 5/2026*
National Sun Yat-Sen University **Kaohsiung, Taiwan**
Bachelor of Science in Applied Mathematics *9/2019 - 6/2023*
***Honors & Awards*:** The Valedictorian of Department of Applied Mathematics, Distinguished Service Award

PROJECT EXPERIENCE

cAIuldron (Recipe Generation LLM) cAIuldron - Kuan-Chen Chen **Gainesville, FL**
- Built v1 end-to-end ML pipeline integrating CLIP (ViT-Base) and DETR (ResNet-50) for multi-ingredient detection (88% F1, 525+ classes); fine-tuned Llama 3.2 1B via QLoRA, training only 0.23% of parameters (2.8M) to achieve 92/100 recipe quality on 7,913 curated RecipeNLG recipes.
- Rebuilt as cloud-native v2 using LangGraph StateGraph orchestration; replaced local GPU models with Groq Llama 4 Scout 17B (vision, ~1s) and Llama 3.3 70B, eliminating 4–6 GB VRAM dependency and upgrading from sequential Python calls to parallel agent execution.
- Integrated ChromaDB RAG + Tavily web search via LangGraph fork/join; delivered 6-tab Gradio UI with 5 concurrent FLUX.1-schnell async image calls and LangSmith observability, cutting pipeline latency from 40–80s (v1) to 10–20s (v2).

Climate Change Innovation Competition, The Ministry of Education (Taiwan) **Taipei, Taiwan**
- Built DARTS campus mobility prediction system integrating multi-source APIs (TDX/PTX), enabling automated 24/7 data pipeline and real-time forecasting of pedestrian, bicycle, and bus demand
- Developed hybrid time-series model (ARIMA-LSTM) with data preprocessing and feature engineering, achieving 86.1% accuracy; delivered actionable insights adopted by university, optimizing bus schedules and reaching national finals (Top 121 teams)

WORK EXPERIENCE

Florida Gators Women's Basketball **Gainesville, FL**
Data Scientist Intern *8/2025 - Present*
- Built an opponent-strength-adjusted player evaluation system (OA-Rating) using XGBoost and SHAP, analyzing 120K+ game-level observations across 5,500+ NCAA players and 350+ teams
- Implemented an Elo-based team rating system ($\rho = 0.83$ vs. NCAA NET) to track real-time team strength and support game preparation
- Developed a production R Shiny dashboard for player comparison and opponent scouting, delivering insights that contributed to team improvement from 16–17 (NET 60) to 18–15 (NET 48)

Bioinformatics Lab, University of Florida **Gainesville, FL**
Student Research Assistant *12/2024 - Present*
- Designed stacking ensemble algorithm combining XGBoost, LightGBM, and Balanced Random Forest with logistic regression meta-learner to classify 10,486 T-cell receptor CDR3 sequence, improving robustness on highly imbalanced data (1:23) and achieving ROC-AUC 0.876
- Developed biologically-informed data augmentation algorithm using BLOSUM62 substitution matrix, generating conservative sequence variants to expand minority class 10× (432 → 4,622) and mitigate severe class imbalance (1:23 → 1:1.5)
- Engineered 42 physicochemical features from protein sequences and optimized decision thresholds under class imbalance, improving precision to 0.716 for high-confidence, interpretable predictions.

Optimized AI Conference **Atlanta, GA**
Volunteer Staff *4/2025, 3/2026*
Glory Integrated Marketing Co. Ltd **Taipei, Taiwan**
Data Analyst Intern *8/2023 - 12/2023*

SKILLS

- Languages: Python, SQL, R | Databases: MySQL, MongoDB, Oracle
- ML/DL: PyTorch, scikit-learn, XGBoost, LightGBM, HuggingFace Transformers, SHAP
- Web & Viz: Flask, React.js, Gradio, Dash, Shiny, Power BI, Tableau